

Python For Data Science

Practical – V

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Class: SYIT	Batch: 1
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PART – B Practical Questions:

5a. Write a python code to demonstrate importing pandas libraries and create data frame object.

INPUT:

```
import pandas as pd

data = {
    "Name": ["Dhruv", "Aryan", "Sahil"],
    "Age": [50, 40, 30],
    "Qualified": [True, False, False]
}
df = pd.DataFrame(data)

print(df)
print("S007 DHRUV")
```

OUTPUT:

```
===== RESTART: C:/Users/
   Name  Age  Qualified
0  Dhruv   50        True
1  Aryan   40        False
2  Sahil   30        False
S007 DHRUV
.
```

5b. Write a python code to show statistical information on given data set.

INPUT:

```

import statistics

print("Mean:")
print(statistics.mean([1, 3, 5, 7, 9, 11, 13]))

print("\nMedian:")
print(statistics.median([1, 3, 5, 7, 9, 11, 13]))

print("\nMode:")
print(statistics.mode(['red', 'green', 'blue', 'red']))

print("\nStandard Deviation:")
print(statistics.stdev([1, 3, 5, 7, 9, 11]))

print("\nVariance:")
print(statistics.variance([1, 3, 5, 7, 9, 11]))

print("S007 DHRUV")

```

OUTPUT:

```

===== RESTART: C:/Users
Mean:
7

Median:
7

Mode:
red

Standard Deviation:
3.7416573867739413

Variance:
14
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```

5c. Write a python code to create pandas series from dictionaries.

INPUT:

```

import pandas as pd

dictionary = {'A': 10, 'B': 20, 'C': 30}

series = pd.Series(dictionary)

print(series)
print("S007 DHRUV")

```

OUTPUT:

===== RESTART:

```
A    10
B    20
C    30
dtype: int64
S007 DHRUV
```

INPUT:

```
import pandas as pd

dictionary = {'A': 50, 'B': 10, 'C': 80}

series = pd.Series(dictionary, index=['B', 'C', 'A'])

print(series)
print("S007 DHRUV")
```

OUTPUT:

===== RESTART:

```
B    10
C    80
A    50
dtype: int64
S007 DHRUV
```

5d. Write a python code to demonstrate filter pandas series with Boolean arrays.

INPUT:

```
import pandas as pd

dict = {'Name':["DHRUV", "SAHIL", "ARYAN", "KAIF"],
        'Degree': ["MBA", "BCA", "M.Tech", "MBA"],
        'Age': [18, 19, 21, 20]}

df = pd.DataFrame(dict, index = [True, False, True, False])

print(df)
print("S007 DHRUV")
```

OUTPUT:

```
===== RESTART: C:/Users/j:
      Name  Degree  Age
True  DHRUV   MBA   18
False SAHIL   BCA   19
True  ARYAN  M.Tech  21
False KAIF   MBA   20
S007 DHRUV
```